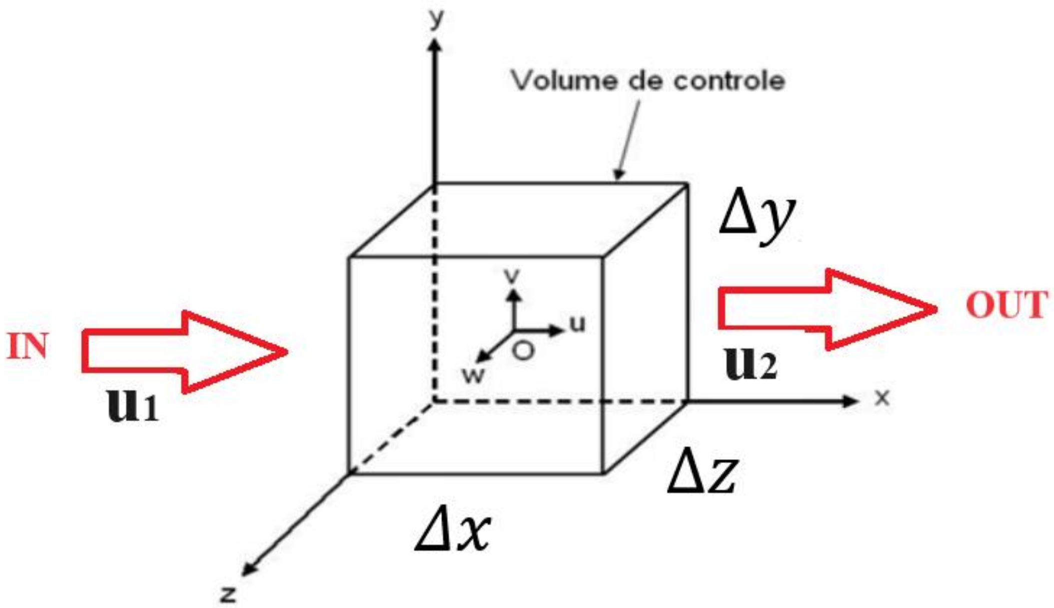
Leis de Conservação da Mecânica dos Fluidos aplicadas à Hidrodinâmica do Navio

I – Lei da Conservação de Massa

Vol . .

p τ - -

. . x



p_{rxw} p_{l} p_{xer}

-

p $\tau x D$

x $\frac{}{w}$ D C -

p_{rxw} τ x D

p_{l} τ x D

- -

p_{xer} $\frac{p}{w}$ C $\frac{\tau}{w}$ - -

p_{rxw} p_{l} p_{xer}

$$\tau \quad x \quad D \qquad \tau \quad x \quad D \qquad \frac{\tau}{w} \quad - \quad -$$

$$- \quad -$$

$$D \qquad D \qquad -$$

$$\tau \quad x \qquad \tau \quad x \qquad - \qquad \frac{\tau}{w} \quad - \quad -$$

$$\tau \quad x \qquad \tau \quad x \qquad \tau x$$

$$\tau x \quad - \qquad \frac{\tau}{w} \quad - \quad -$$

$$\tau x \quad - \quad \mathrm{C} \quad \tau y \quad - \quad \mathrm{C} \quad \tau z \quad - \qquad \frac{\tau}{w} \quad - \quad -$$

$$y \qquad z$$

$$- \quad -$$

$$\frac{\tau x}{} \quad \mathrm{C}\frac{\tau y}{\mathrm{C}} \quad \mathrm{C}\frac{\tau z}{} \qquad \frac{\tau}{w}$$

$$. \qquad \frac{-}{-} \qquad \frac{-}{-}\mathrm{C}$$

$\overline{\quad \tau x \quad}$ C $\overline{\quad \tau y \quad}$ $\overline{\quad \tau z \quad}$ $\frac{\tau}{w}$

τ $\frac{\tau}{w}$

— C— —

u, v e w *x, y e z*

τ - g_D $\frac{g}{gw}$ τCg $_{ro}$

$\frac{\tau}{w}$

τ

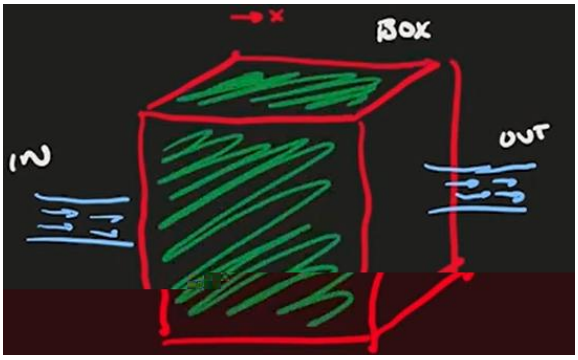
$\overline{\quad \tau x \quad}$ C $\overline{\quad \tau y \quad}$ $\overline{\quad \tau z \quad}$ τ $\overline{x}$ C $\overline{y}$ $\overline{z}$

$\overline{x}$ C $\overline{y}$ $\overline{z}$

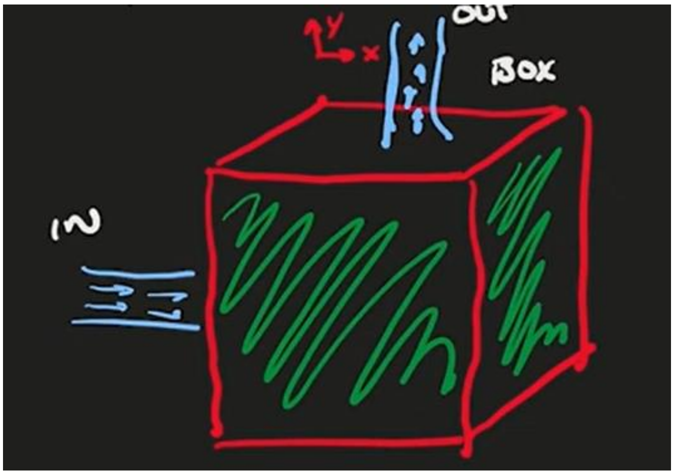
-

$\underline{x}$

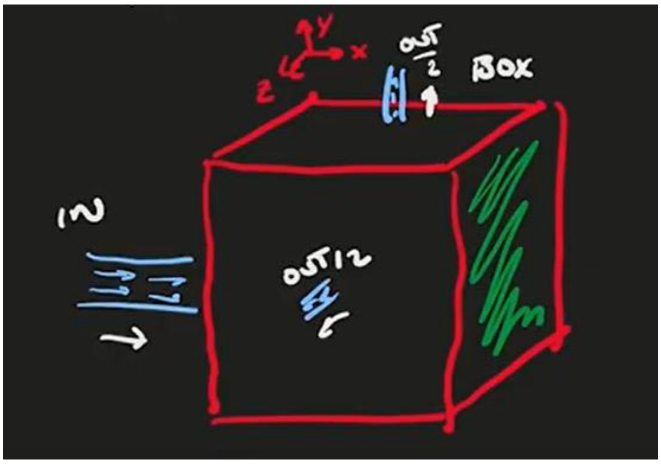
$\underline{y}$ $\underline{z}$



$\underline{x}$



$C\underline{x}$ $C\underline{y}$



$C\frac{x}{}$ $C\frac{y}{}$ $\frac{z}{}$

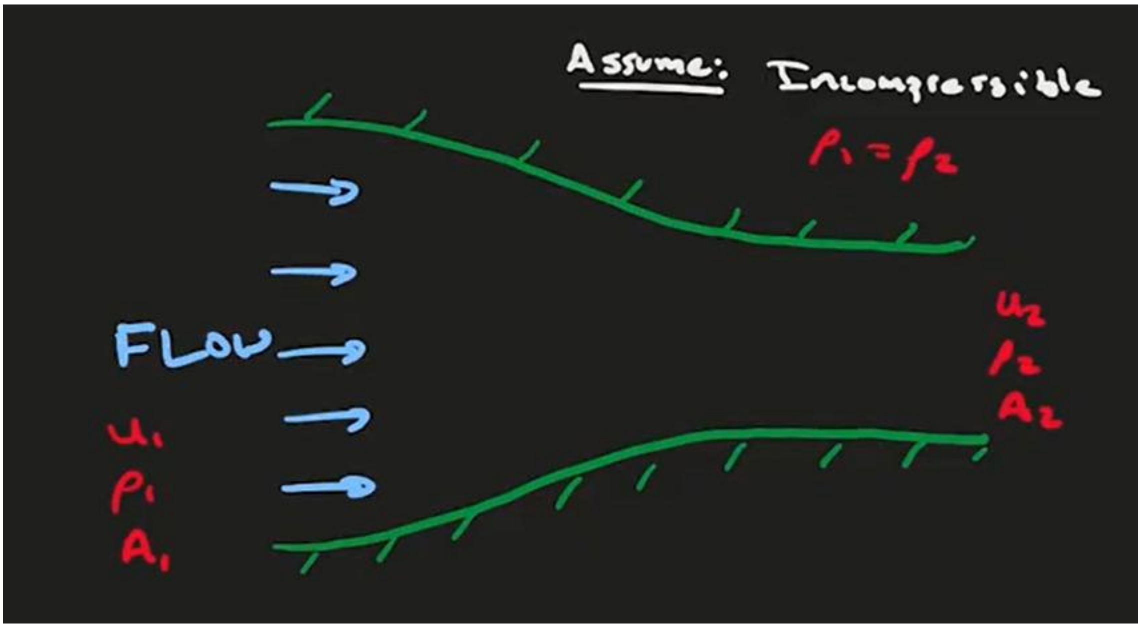
II – A lei de conservação da quantidade de movimento

$$\frac{p}{w} \quad C$$

A derivada material, derivada substantiva ou derivada total

$$\tau \frac{x}{w}$$

(u e u) τ hGτ *(A e A)*



τ x D τ x D

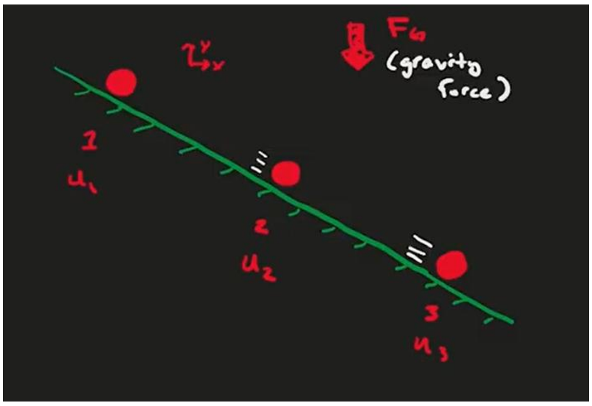
τ $C\tau$

x D x D

D D x x

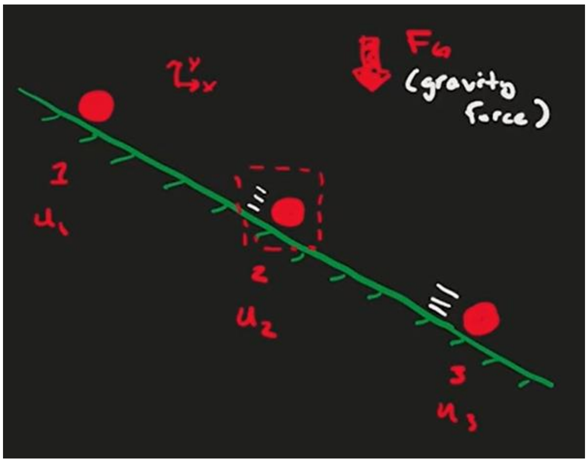
$\frac{x}{w}$

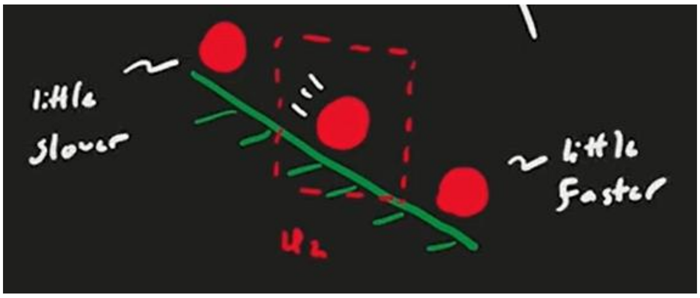
$\frac{x}{w}$



x x x


u
t
$a = \frac{\Delta u}{\Delta t}$


F_G
(gravity force)
1
u_1
2
u_2
3
u_3


~ little slower
~ little faster
u_2

$$\frac{x}{w} \quad \frac{x}{w}\frac{}{} \quad \frac{}{w}\frac{x}{} \quad x\frac{x}{}$$

$$\frac{x}{w} \quad \frac{x}{w}\frac{}{} \quad \frac{}{w}\frac{x}{} \quad y\frac{x}{}$$

C

$$d \quad \frac{x}{w} \quad x\frac{x}{} \quad y\frac{x}{} \quad z\frac{x}{}C$$

DvCdfhchud hvCqdvCgluh hvĆdChĈdCvhu rCuhvs hf lydp hq hC

$$d \quad \frac{}{w} \quad x\frac{y}{} \quad y\frac{y}{} \quad z\frac{y}{}\text{C}$$

C

$$d \quad \frac{z}{w} \quad x\frac{z}{} \quad y\frac{z}{} \quad z\frac{z}{}\text{C}$$

C

dvrCrChvfrdp hq rCvhmdCshup dqhq h.Chp vhC

$$\text{C}\frac{x}{w} \quad .\text{C}\frac{y}{w} \quad \text{ChC}\frac{z}{w}$$

Forças atuantes em um elemento fluido

$$\frac{\frac{p}{o}}{w} \quad \text{C}\frac{}{o}$$

$$\frac{p}{ro} \quad \tau$$

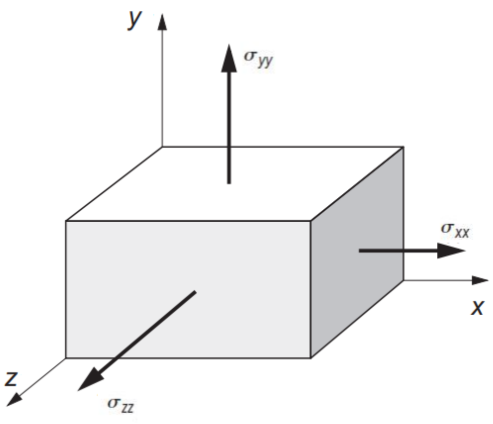
$$\frac{\tau}{w} \quad \text{C}\frac{}{o}$$

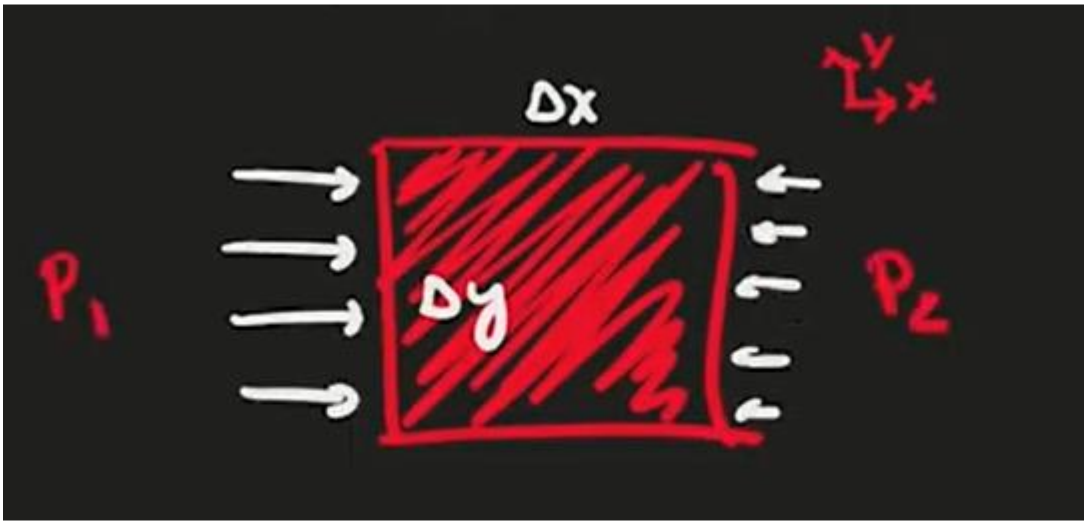
$$\tau\frac{}{w} \quad \text{C}\frac{}{o}$$

As forças normais ou de pressão

lm i

j





s C $\overline{D}$ C s D

s D s D

D D

s s

s s s

s

$$\underline{\qquad\qquad} \quad \frac{s}{}$$

$$\frac{}{ro} \quad \frac{s}{}$$

$$\frac{}{ro} \quad \frac{s}{}$$

$$\frac{}{ro} \quad \frac{s}{}$$

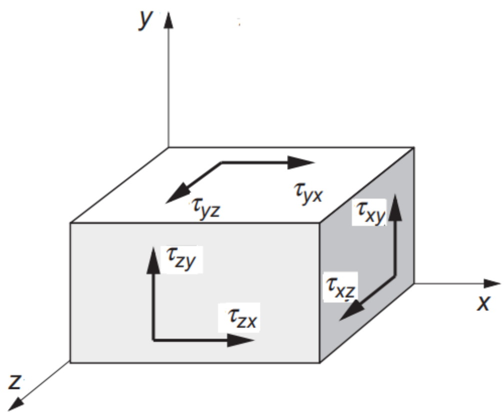
$$\tau \frac{x}{w} \qquad \frac{s}{}$$

$$\tau \frac{y}{w} \qquad \frac{s}{}$$

$$\tau \frac{z}{w} \qquad \frac{s}{}$$

As forças tangenciais, também conhecidas como forças friccionais ou viscosas

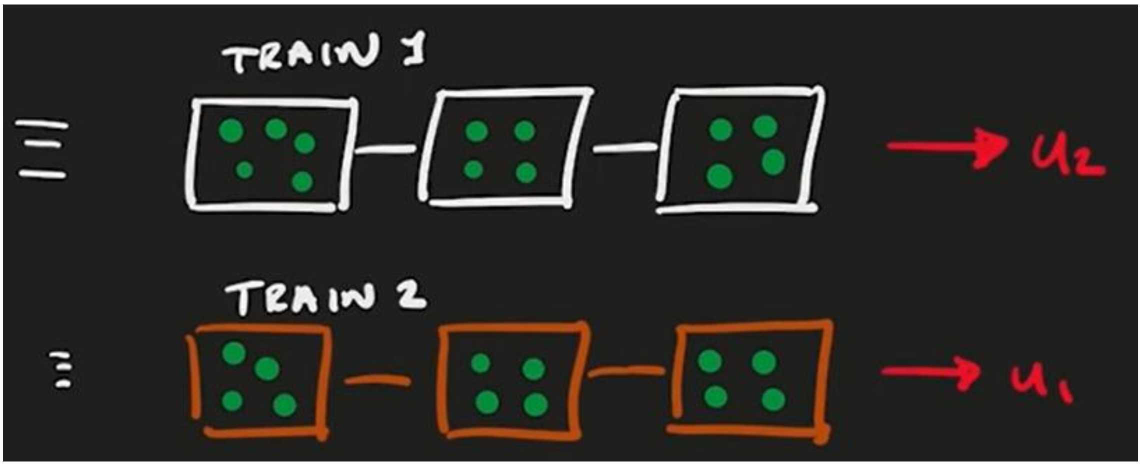
i $\qquad$ lm j

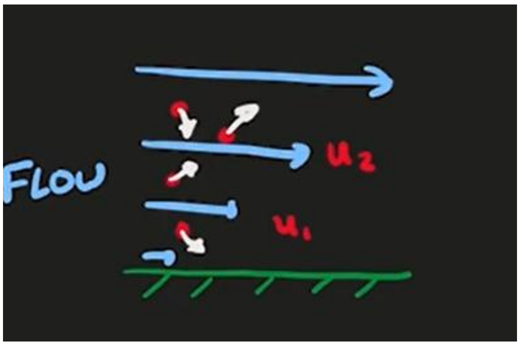


$x \quad x$

x

x





C_D C D

$C\frac{}{D} \quad \frac{p}{D}\frac{x}{w}$

$\frac{x}{w}$

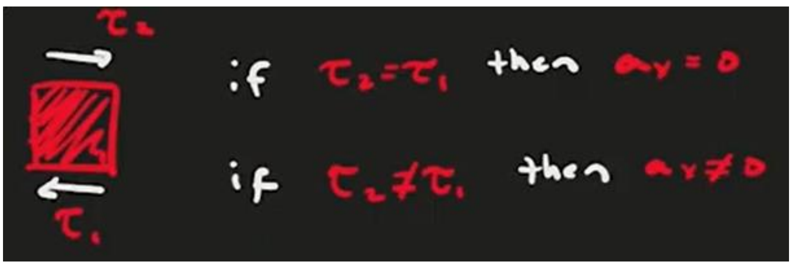
$\frac{x}{w} \quad \frac{x}{w} \quad \frac{}{w}\frac{x}{}$

$\frac{p}{D}\frac{x}{w} \quad \frac{p}{D}\frac{}{w}\frac{x}{}$

$\frac{p}{D}\frac{}{w}$

$\frac{p}{D}\frac{}{w}$

$\frac{x}{}$



D D

D D

$\frac{x}{}$

$\frac{x}{}$ $\frac{x}{}$

$\frac{}{o}$ $\frac{x}{}$

$\frac{}{o}$ $\frac{x}{}$ $\frac{x}{}$ $\frac{x}{}$

$\tau \frac{x}{w}$ $\frac{s}{}$ $\frac{x}{}$ $\frac{x}{}$ $\frac{x}{}$

$\tau \frac{y}{w} \quad \frac{s}{} \quad \frac{y}{} \quad \frac{y}{} \quad \frac{y}{}$

$\tau \frac{z}{w} \quad \frac{s}{} \quad \frac{z}{} \quad \frac{z}{} \quad \frac{z}{}$

As forças de corpo

Gp Gτ

———— Gτ

$\tau \frac{x}{w} \quad \frac{s}{} \quad \frac{x}{} \quad \frac{x}{} \quad \frac{x}{} \quad \tau$

$\tau \frac{x}{w} \quad x\frac{x}{} \quad y\frac{x}{} \quad z\frac{x}{} \quad \frac{s}{} \quad \frac{x}{} \quad \frac{x}{} \quad \frac{x}{} \quad \tau \quad C$

hGrup dGvlp lodu Gqdv Gluh hvĆdĆhĈdĈvhu rĈuhvshf lydp hq hC

$\tau \frac{}{w} \quad x\frac{y}{} \quad y\frac{y}{} \quad z\frac{y}{} \quad \frac{s}{} \quad \frac{y}{} \quad \frac{y}{} \quad \frac{y}{} \quad \tau \quad C$

C

$\tau \frac{z}{w} \quad x\frac{z}{} \quad y\frac{z}{} \quad z\frac{z}{} \quad \frac{s}{} \quad \frac{z}{} \quad \frac{z}{} \quad \frac{z}{} \quad \tau \quad C$

C

DvC u vChtxd hvCdflp dCfruuhvsrqghp CdCdp rvdC tx rCghCQdylhu VrnhvCtxhCqdC irup dCyh ruldC Cdsuhvhq dgdCsruC

τ —— s τ C
w

C

C CDCChlCghCfrqvhuyd rCghChqhujldC

Srgh vhCglylgluCdChqhujldCghCxp Cvlv hp dChp CgxdvCfodvvhvCglv lq dv-CdChqhujldC dup dCdhqdgdChCdChqhujldChp C udqvl r-CDCsulp hludChv Cuhodflrqdgd Cgluh dp hq hC uhodflrqdgdCfrp CdCp dvvdCgrCsu sulrCvlv hp d.ChChqyroyhCdvChqhujldvCq huqd.Cflq lfdC hCsr hqfldoCdCvhjxqgdCfodvvhCvhCuhodflrqdCfrp CdChqhujldC urfdgdChq uhCvlv hp dv.C lqfoxlqgrCrCfdoruChCrC udeddkr-C rqvlghudqgrC CdChqhujldCr doCdup dCdhqdgdCsruCxp C vlv hp dCtxh.Cqxp Cq huydorCghChp sr.CuhfhehxCxp dCtxdq lgdghCghCfdoruC ChCh hfx rxC xp C udeddkrC .CdCfrqvhuyd rCghvvhCvlv hp dC CudgxCdlgdCd udy vCgdCh suhvv r-C

C

DCh suhvv rCdflp dC CdC'sulp hludCchlCgdChup rglq p lfd-CC

q r.CdC d dCghCyduld rCgdChqhujldCsruCxqlgdghCghC hp srC C$\frac{g}{gw}$ .CdvC urfdvCghC μqhujldChih xdgdvCfrp CrCh hulruCvreCrup dCghCfdoruC$\frac{\mu}{gw}$ .ChCdCghC udeddkr.C$\frac{g}{gw}$ Cuhvxodqgr Chp -C

$\frac{g}{gw}$ C

CtxhCvljqlCfdCglCdhuCtxhCdCd dCghChq udgd.CsruCxqlgdghCghChp sr.CghChqhujldC up lfdC hCp hf qlfdCqxp Cvlv hp dCp hqrvCdCfruuhvsrqghq hCd dCghCvd gdCghyhu Cj xdoduCdCd dC ghCdfxp xod rCghChqhujldCqrCq hulruCghvvhCp hvp rCvlv hp dC

SdudCuhodflrqduChv hvCfrqfhl rvCdflp dCfrp CdCqr rCghCyroxp hCghCfrq urch.Chp suhjd vhCrCfrqfhl rCghCghqvlgdghCp vvlfdCghChqhujldCdup CdhqdgdCs horCvlv hp d.Ch-C

h

μ

p

$\frac{g}{gw}$ $\frac{}{w}$ $h\tau Cgy$ o $h\tau$ - gDC

p

h

p h

h

h suhvv r ylv rvdv

h

p

suhvv r

gD

g suhvv r s - gD

suhvv r s - gD

$ylv\ rvdv$

$g\ \ _{ylv\ rvdv}\ \ -\ \ gD$

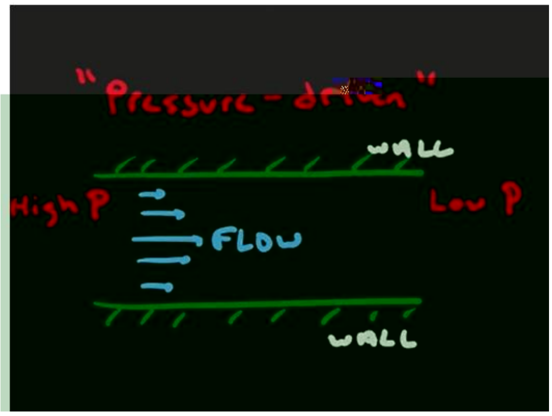
$_{ylv\ rvdv}\ \ -\ \ gD$

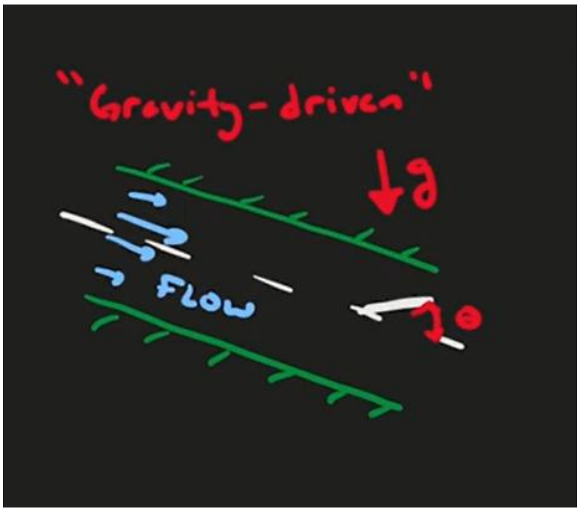
$_{p}\ \ s\ \ -\ \ gD\ \ -\ \ gD$

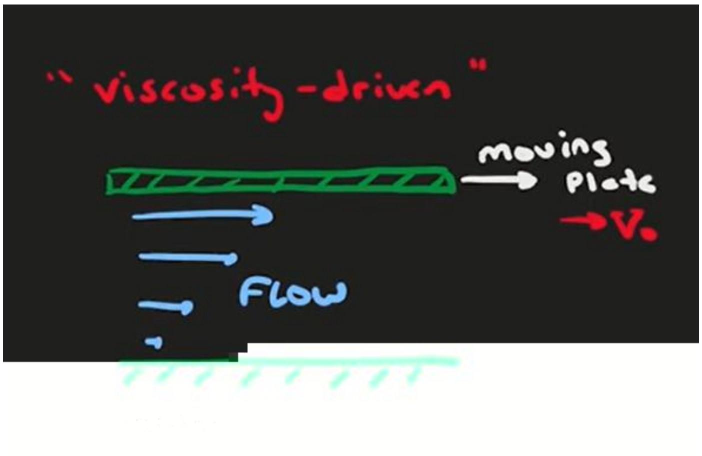
$\overline{}\ w\ \ _{ro}\ \ x\ \ -\ \ \tau Cgy\ \ o\ \ _{V}\ \ x\ \ -\ \ s\ \ -\ \ gD$

Hipóteses comum na mecânica dos fluidos

Aplicações simples das leis de conservação de massa e de conservação de quantidade de movimento - Escoamentos laminares internos fechados


"Pressure-driven"
WALL
High P
FLOW
Low P
WALL

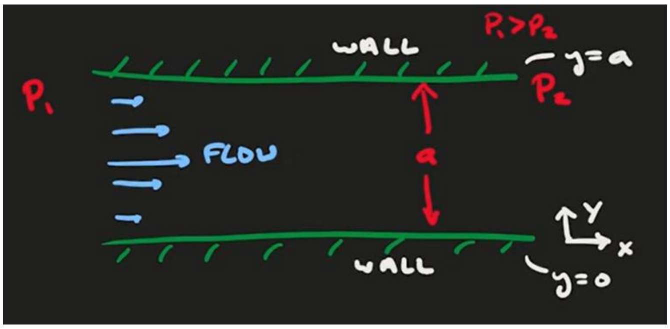




Escoamento plenamente desenvolvido dominado pelas forças de pressão

s s s s

d d



τ vw

$\frac{-}{w}$

$\frac{x}{}$

$\frac{-}{}$ z

τ

x 3 x d

Gz 3 z d

y 3 y d

$\frac{x}{}$ G$\frac{y}{}$ $\frac{z}{}$

$\frac{x}{}$ $\frac{z}{}$

$\frac{y}{}$

$\frac{y}{}g \qquad y \qquad vw \quad F$

F

$x \quad 3 \qquad x \quad d$

$x \quad 3 \quad \frac{s}{} \qquad F \qquad F$

F

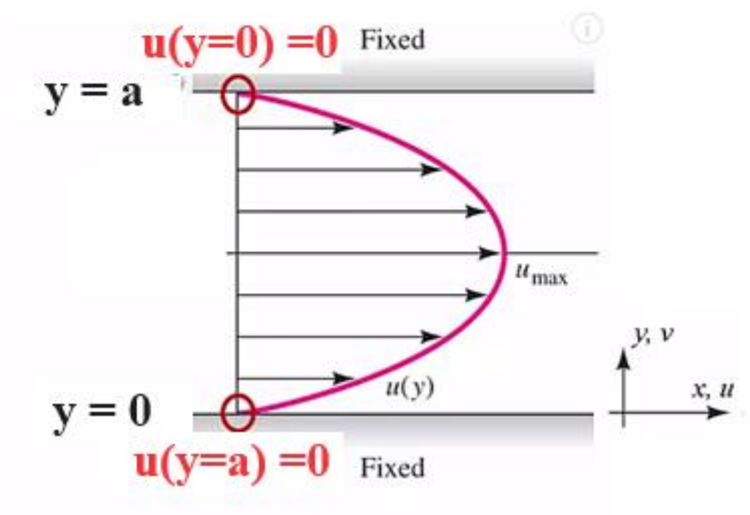
$x \quad d \quad \frac{s}{} d \qquad F \; d$

$F \qquad \frac{s\;d}{}$

$F \quad F$

$x \quad \frac{s}{} \qquad \frac{s\;d}{}$

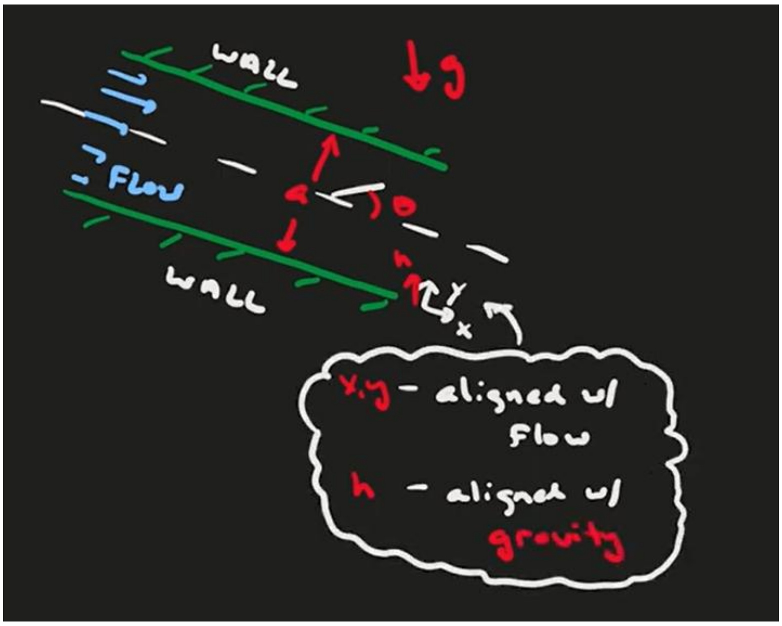
$x \quad \frac{s}{} \qquad d$



Escoamento plenamente desenvolvido dominado pela força de corpo

d

d



τ vw

$\frac{-}{w}$

$\frac{x}{}$

$\frac{-}{}$ z

x 3 x d

$\mathbb{C}z$ 3 z d

y 3 y d

$\frac{x}{}$ C $\frac{y}{}$ $\frac{z}{}$

$\frac{x}{}$ $\frac{z}{}$

$\frac{y}{}$

$\frac{y}{}g$ y vw F

y $_{3.d}$

F y

τ $\frac{x}{w}$ $x\frac{x}{}$ $y\frac{x}{}$ $z\frac{x}{}$ $\frac{s}{}$ $\frac{x}{}$ $\frac{x}{}$ $\frac{x}{}$ τ C

$\frac{x}{w}$

$x\frac{x}{}$ $x\frac{x}{}$

$z\frac{x}{}$ $\frac{x}{}$

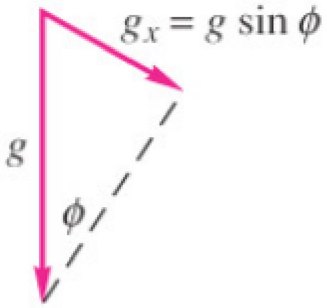
τ

y $y\frac{x}{}$

$\frac{s}{}$ $\frac{x}{}$ τ

$\frac{x}{}$ $\frac{s}{}$ τ

vlq



vlq $\frac{g}{g}$

$\frac{x}{} \quad -\frac{s \quad \tau}{} \quad vw$

$\frac{x}{} \quad -\frac{s \quad \tau}{} \quad F$

x

$x \quad -\frac{s \quad \tau}{} \quad F \quad F$

$x \quad 3 \quad x \quad d$

$x \quad 3 \quad -\frac{s \quad \tau j}{} \quad F \quad F$

F

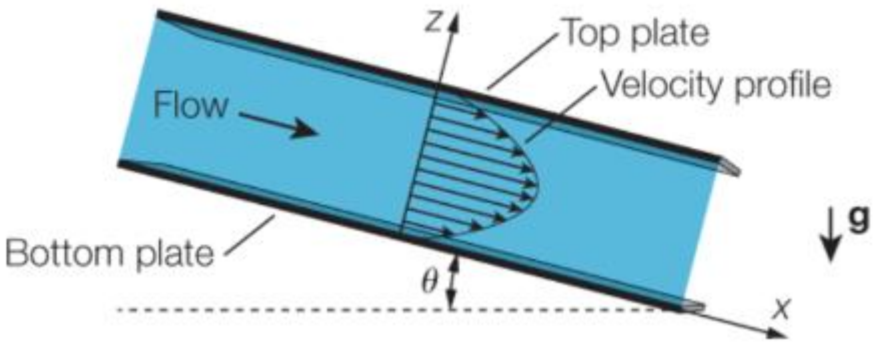
$x \quad _d \quad -\frac{s \quad \tau j}{\quad} \frac{d}{\quad} \quad F \; d$

$F \quad -\frac{s \quad \tau j}{\quad} \frac{d}{\quad}$

$F \quad F$

$x \quad \frac{}{} \frac{s \quad \tau}{\quad} \quad -\frac{s \quad \tau}{\quad} \frac{d}{\quad}$

$x \quad \frac{}{} \frac{s \quad \tau}{\quad} \quad d$



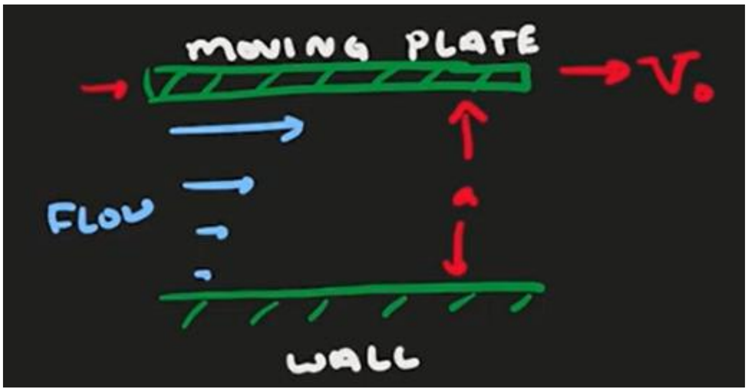
Escoamento plenamente desenvolvido dominado pelas forças friccionais

V s

s s s

d

d



τ vw

$\frac{-}{w}$

$\frac{x}{}$

$\frac{-}{}$ z

x 3 x d 3

Cz 3 z d

y 3 y d

$\frac{x}{}$ $C\frac{y}{}$ $\frac{z}{}$

$\frac{x}{}$ $\frac{z}{}$

$\frac{y}{}$

$\frac{y}{}g$ y vw F

$y \quad {}_{3.d}$

$F \qquad y$

$\tau \quad \frac{x}{w} \quad x\frac{x}{} \quad y\frac{x}{} \quad z\frac{x}{} \quad \frac{s}{} \quad \frac{x}{} \quad \frac{x}{} \quad \frac{x}{} \quad \tau \quad C$

$\frac{x}{w}$

$x\frac{x}{} \qquad x\frac{x}{}$

$z\frac{x}{} \qquad \frac{x}{}$

τ

$y \qquad y\frac{x}{}$

$\frac{x}{}$

$\frac{x}{} \quad F$

x

$x \quad F \quad F$

$x \quad {}_{3} \qquad x \quad {}_{d} \quad {}_{3}$

$x \quad {}_{3} \quad F \quad F$

F

$x \quad {}_{d} \quad F \ d \quad {}_{3}$

$F \quad \frac{}{d}$

$x \quad \frac{3}{d}$ ((((((C d